byteME

MINDS IN ENGINEERING



N = 12 SPINS · T = 2

QISKIT FALL FEST 2026 · TRACK 3

ISING CHAIN ON A NOISY QUANTUM SIMULATOR

Trotter dynamics, noise and error mitigation

Level reached: Beginner and Intermediate

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github.com/SnehilK3372/qiskit-fall-fest-2026-IIITD

SUMMARY

01

THE QUESTION

How well can a quantum computer simulate a chain of tiny magnets (the transverse-field Ising model) as it changes over time, and what happens when real-world noise is added? We score every method against the exact answer.

02

WHAT WE DID

Beginner: 8 magnets on a perfect simulator, comparing two ways of slicing time (Lie-Trotter and Suzuki) with the exact answer. Intermediate: 12 magnets with noise copied from IBM's Fez chip, then fixed with readout correction and zero-noise extrapolation (ZNE).

03

WHAT WE FOUND

Beginner: Lie-Trotter is cheaper for Mz (168 vs 314 CX), Suzuki is cheaper for Mx (314 vs 2240 CX). Intermediate: readout correction + exponential ZNE cut the error from 0.0308 to 0.0164 (Mz) and from 0.0850 to 0.0160 (Mx); no method was worse than raw.

8 → 12
spins, Beginner to Intermediate

0.01
error target (Beginner)

1 · 3
ZNE noise levels

0
real QPU runs

APPROACH: BEGINNER

01
Hamiltonian
H = −J ΣZZ − h ΣX, open chain, SparsePauliOp

→

02
Product formula
LieTrotter (1st) and SuzukiTrotter (2nd), dt = T/n

→

03
Simulate
Exact Statevector, state kept after every step

→

04
Compare
Mz, Mx, ⟨Z0 Zr⟩ vs scipy expm

→

05
Cost
Transpile to rz, sx, x, cx; count CX

SETUP

8 spins, J = h = 1, start |00000000⟩, evolve to T = 2. Circuits are decomposed so Trotter error is really present. Mz and Mx are the mean ⟨Z⟩ and ⟨X⟩ per spin; our helpers match StatevectorEstimator (max difference 4.7e-16).

EXTRA ANALYSIS

Correlations vs distance and time (4 and 8 spins), error scaling with dt, accuracy vs depth and CX count, Suzuki order 4. Fitted error slopes are about 1, 2 and 4 for orders 1, 2 and 4.

RESULTS: BEGINNER

Formula	Observable	Largest dt	Steps n	CX gates	Max error
Lie-Trotter (1st)	Mz	0.1667	12	168	0.0099
Lie-Trotter (1st)	Mx	0.0125	160	2240	0.0082
Suzuki (2nd)	Mz	0.1667	12	314	0.0099
Suzuki (2nd)	Mx	0.1667	12	314	0.0075

> Mz: both formulas give identical error at the same n, so Lie-Trotter is cheaper (168 vs 314 CX at n = 12). Mx: Suzuki needs 12 steps instead of 160 and 314 CX instead of 2240, so it is far cheaper overall. Mz commutes with the ZZ terms and the start state is their eigenstate, which hides the first-order error; Mx does not, so it shows the error in full.

Largest dt with max error ≤ 0.01 over t ≤ 2, 8 spins. The last column is the actual worst-case error at that dt. CX counts are after transpiling to rz, sx, x, cx.

APPROACH: INTERMEDIATE

01
Trotter circuit
Mz: 1st order · Mx: 2nd order

→

02
Fold
noise factors 1 and 3 (U U†U)

→

03
FakeFez noise
Aer, optimization_level=3

→

04
Readout fix
invert measured flips

→

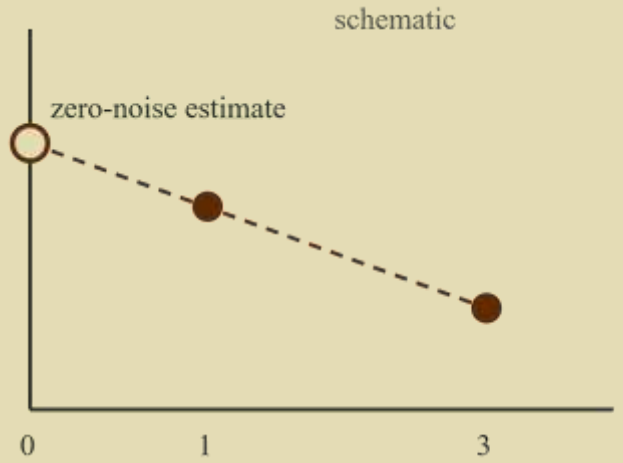
05
ZNE
linear and exponential fits

SETTINGS

N = 12, T = 2, dt = 0.25 (8 steps): fewer, bigger steps mean less noise. Noisy points at t = 0.5, 1.0, 1.5, 2.0, 2000 shots each. Lowest-error 12-qubit line of FakeFez.

CHEAPER CIRCUITS

Mz is identical at 1st and 2nd order, with about half the 2-qubit gates (176 vs 322 at k = 8). Mean readout flips: 0.63% and 0.71%.



ZNE idea: measure at noise 1 and 3, extrapolate to 0

RESULTS: INTERMEDIATE

Method (mean |error| over t = 0.5 to 2.0)	Mz error	Mx error
Noiseless Trotter	0.0146	0.0082
Noisy (raw)	0.0308	0.0850
+ Readout correction	0.0285	0.0805
+ Readout + ZNE linear	0.0175	0.0309
+ Readout + ZNE exponential	0.0164	0.0160

> Depth reduction (k = 8, Mz): using order 1 instead of order 2 cuts 2-qubit gates from 322 to 176 and depth from 1083 to 194; level 3 alone trims order 2 from 338 to 322 gates. Readout + exponential ZNE cut the error from 0.0308 to 0.0164 (Mz) and from 0.0850 to 0.0160 (Mx), close to the noiseless floor. No method was worse than raw noisy. Linear ZNE helped less on Mx (0.0309), likely because the decay is curved; the remaining Mz error is mostly Trotter error.

HOW WE CHECKED OUR RESULTS

AGAINST REFERENCES

Exact expm (8 spins) and expm_multiply (12 spins, 4096-dim). Helpers matched StatevectorEstimator on 4 spins (max difference 4.7e-16). Error slopes 1, 2, 4 for orders 1, 2, 4.

INTERNAL TESTS

Ideal Aer run of folded circuits matches noiseless Trotter within shot noise (Mz 0.2485 vs 0.2526 at k = 4). Mz identical for orders 1 and 2. Readout and ZNE fits tested on synthetic data.

HARDWARE RUNS

None. All results come from the FakeFez noisy simulator (calibration snapshot in qiskit-ibm-runtime), so there are no job IDs or QPU seconds to report.

1 · 2 · 4

error slopes matched theory

0

real QPU jobs

2

noise factors in ZNE

LIMITATIONS, RUN, CREDITS

LIMITATIONS

Noise is a model, not the device. No error bars (about ±0.01 at 2000 shots, amplified by ZNE). 4 time points, 2 noise factors; the two-point exponential fit is fixed to decay to 0. Remaining Mz error (0.0164) is close to the Trotter floor (0.0146). Next: error bars, DD, twirling, a real QPU run.

HOW TO RUN

Python 3.12.9. Tested: qiskit 2.5.2, qiskit-aer 0.17.2, qiskit-ibm-runtime 0.50.0, scipy 1.18.1 (minimum assumed: qiskit ≥ 1.2, aer ≥ 0.15, runtime ≥ 0.30), numpy, matplotlib. Run main.ipynb or intermediate_cells.ipynb top to bottom (intermediate run took about 7.5 min, 435 s; QUICK = True for a smoke test).

CREDITS

Tutorial: BasQ Qiskit Fall Fest 2026, Benjamin Tirado (text CC BY 4.0, code Apache 2.0). AI: Claude (Anthropic) for review, fixes and code. [Edit to match.]